

# AI-compute supply chain: chart pack

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01 to 2026-04, checked against Atlas) and Atlas annual data. Sections 3 and 4 use the AI-compute codes only. Details: docs/data.md, docs/supply-chain-narrative.md, results/.

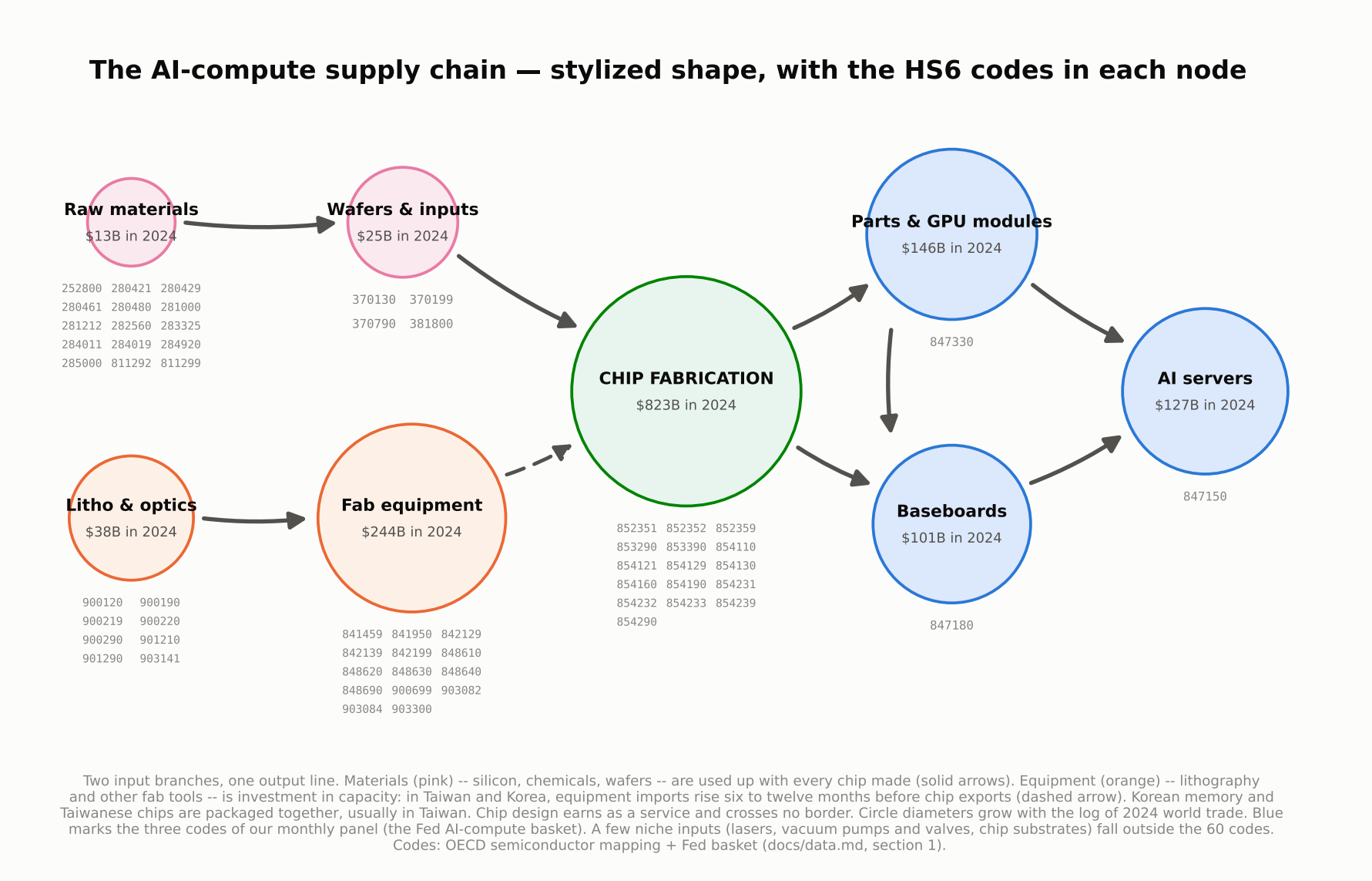
2026-08-20

# 1. The chain as flows

2024 trade flows. China and Hong Kong are counted as one (CHK) throughout. Stages 1-5 use annual Atlas data; stages 6-8 our monthly panel.

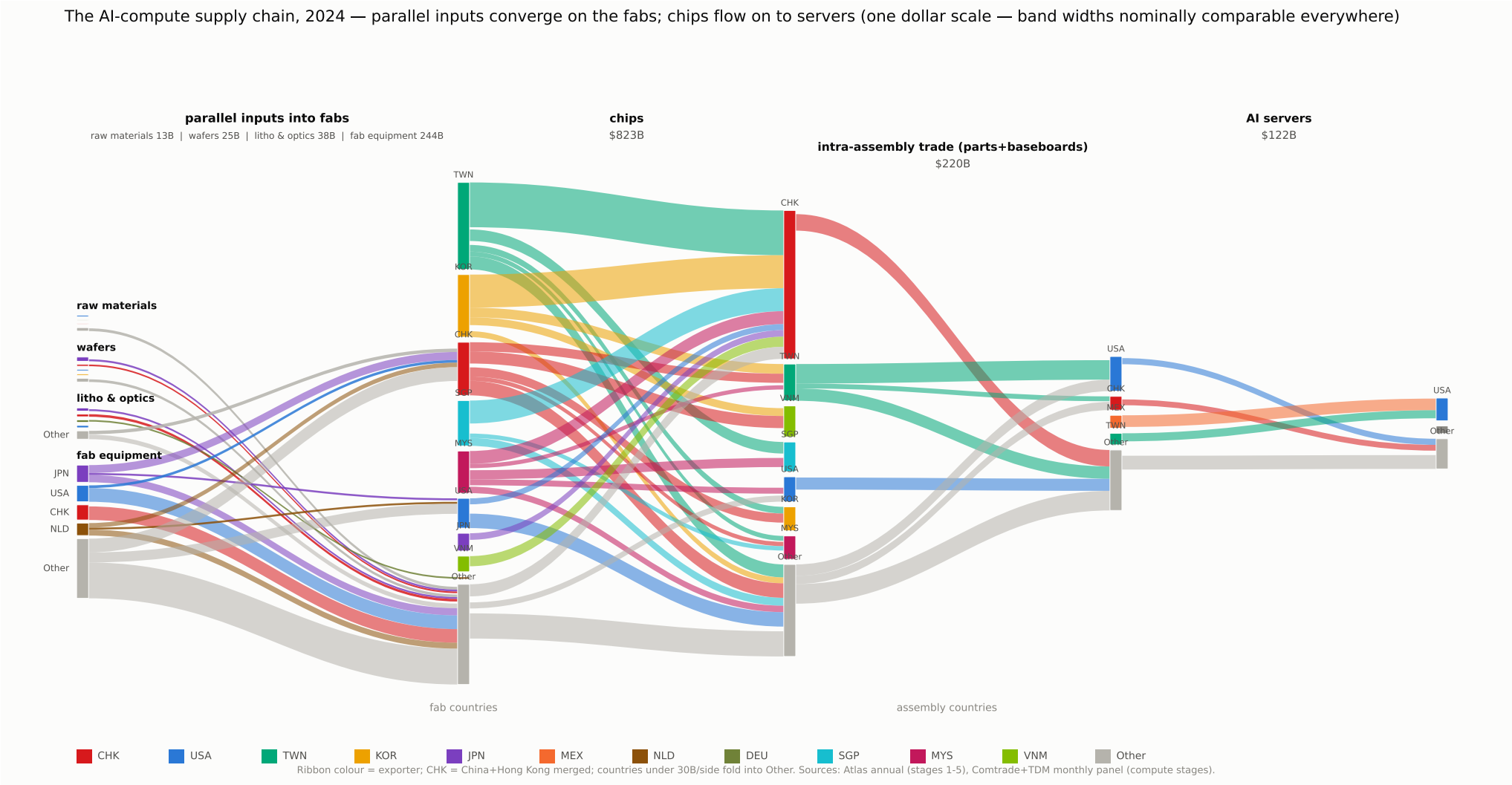
The shape of the chain

Two input branches feed chip making; finished hardware flows out to the right. Pink inputs are used up per chip; orange equipment builds capacity. Blue: the three codes of our monthly panel. Circle size grows with the log of 2024 trade.



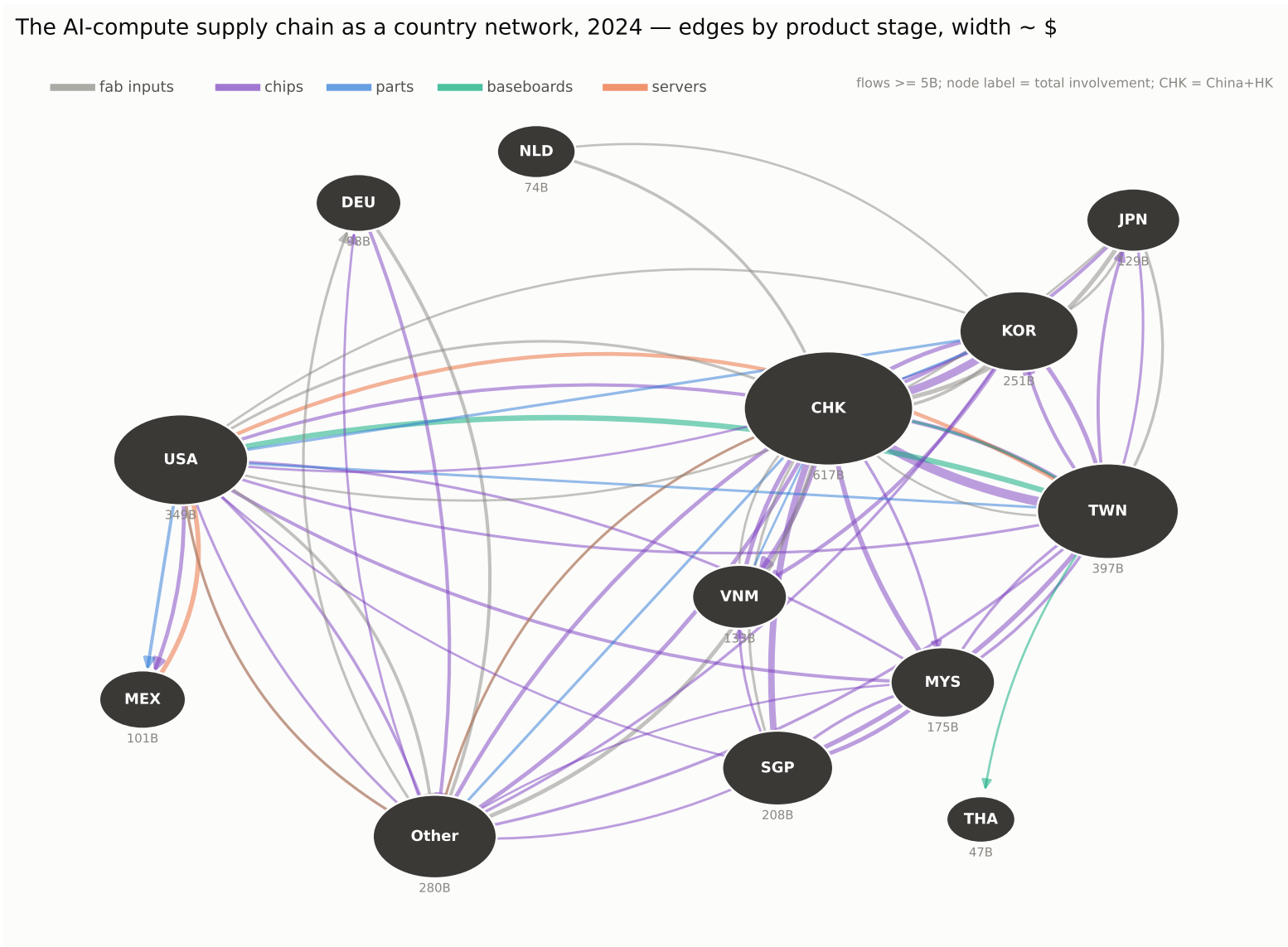
# The whole chain on one dollar scale

Band width = 2024 dollars. The chips stage dominates; small input flows are drawn but too thin to see.



## The same flows as a country map

Countries as nodes, flows as edges colored by stage. Shows combined roles: Mexico takes in chips and parts and ships out servers.

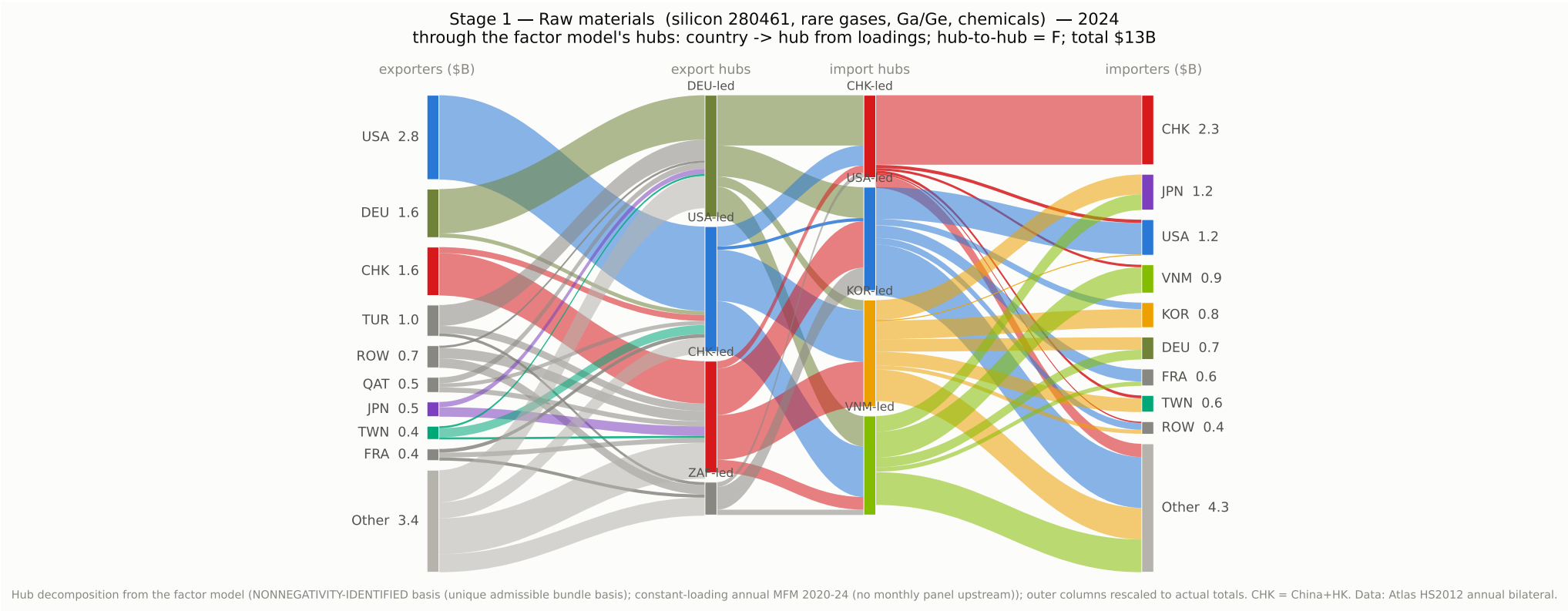


## 2. The chain in stages

One page per stage: exporters on the left, importers on the right, with the model's groupings in between. Groupings fit the 2024 flows closely (R-squared 0.95-0.99).

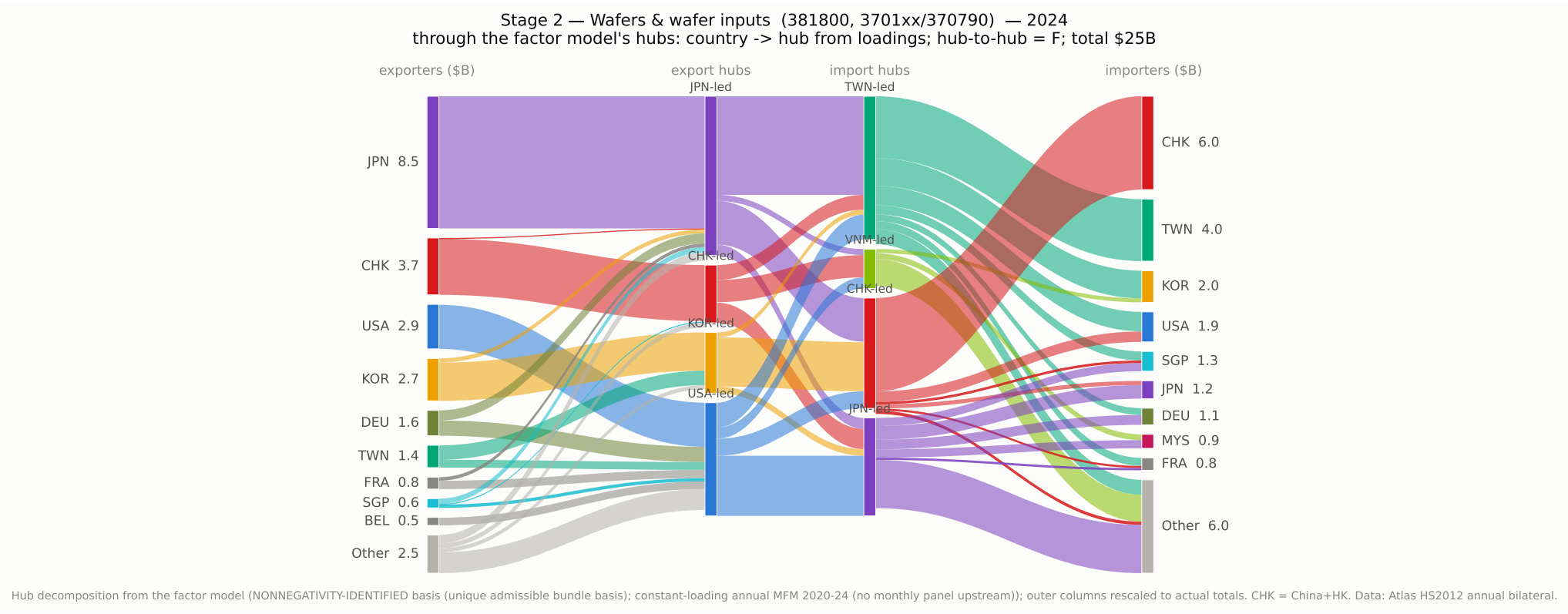
Stage 1 – raw materials

Reading guide for all stage pages: left = exporters, right = importers, middle = the model’s export and import groups. Much raw-materials trade is not chip-related (the codes also cover welding gas, abrasives, fertilizer inputs), so flows scatter widely.



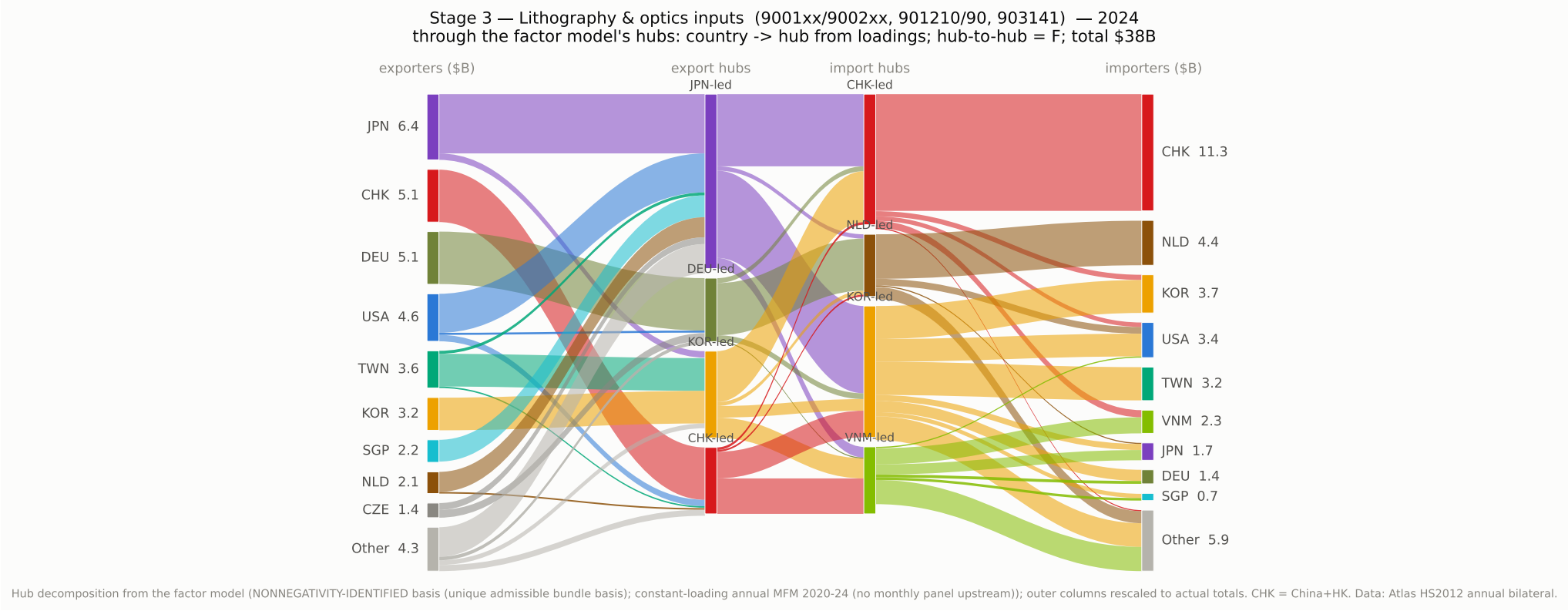
Stage 2 – wafers and wafer inputs

Japan dominates the export side.



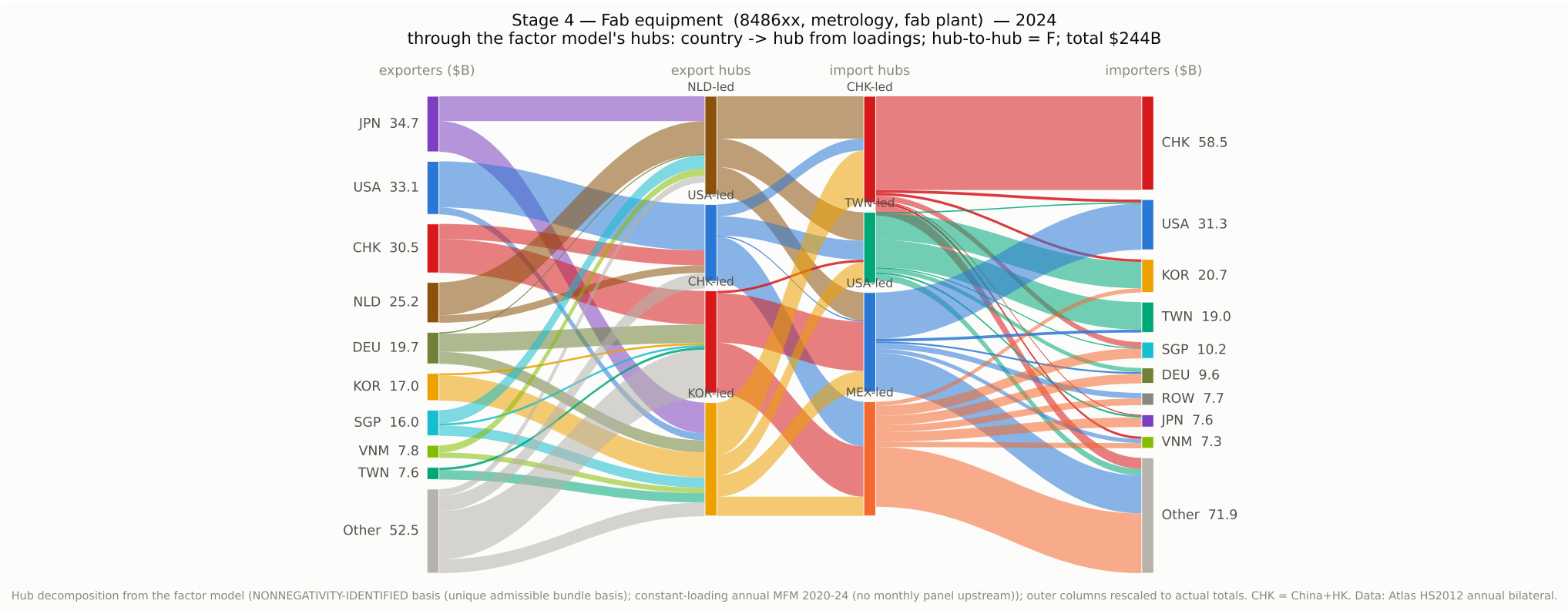


Stage 3 – lithography and optics inputs



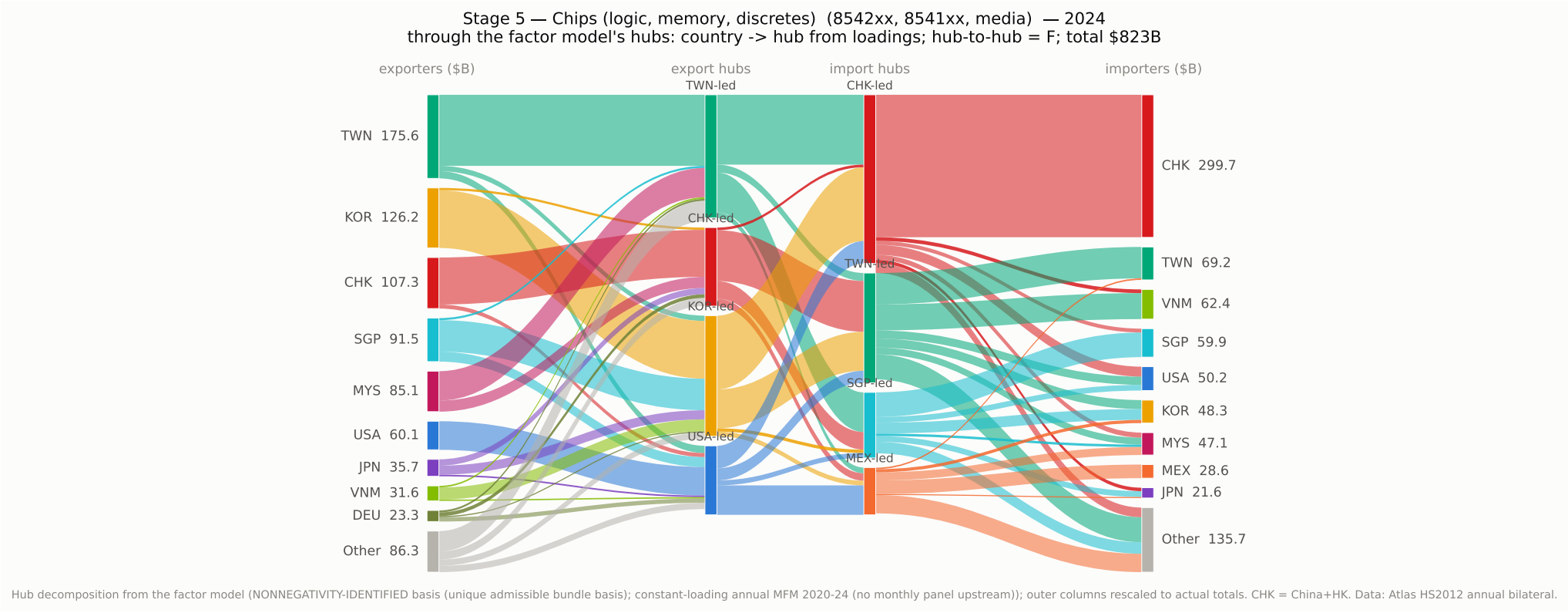
Stage 4 – fab equipment

Machines, not materials: countries buy these to build capacity, and their imports rise 6-12 months before chip exports.

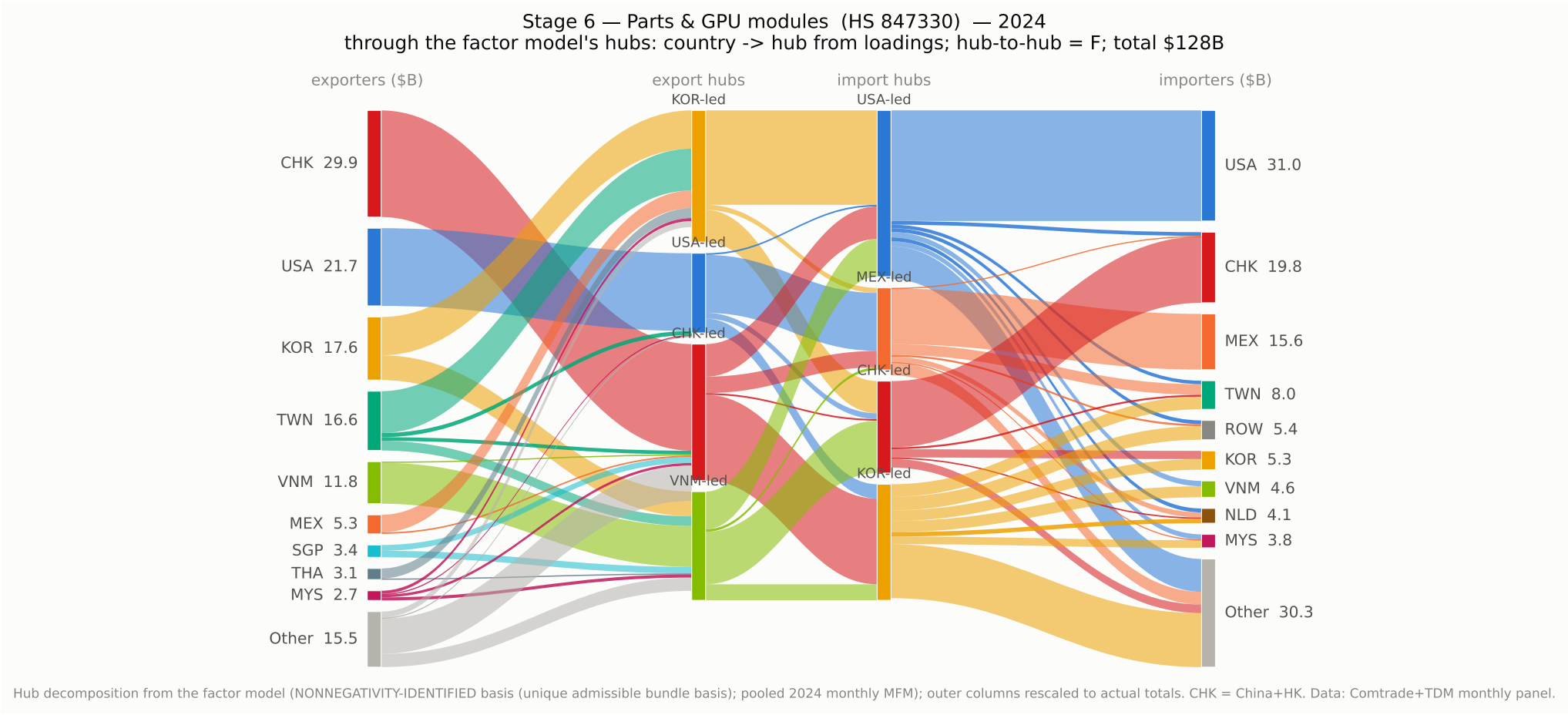


Stage 5 – chips

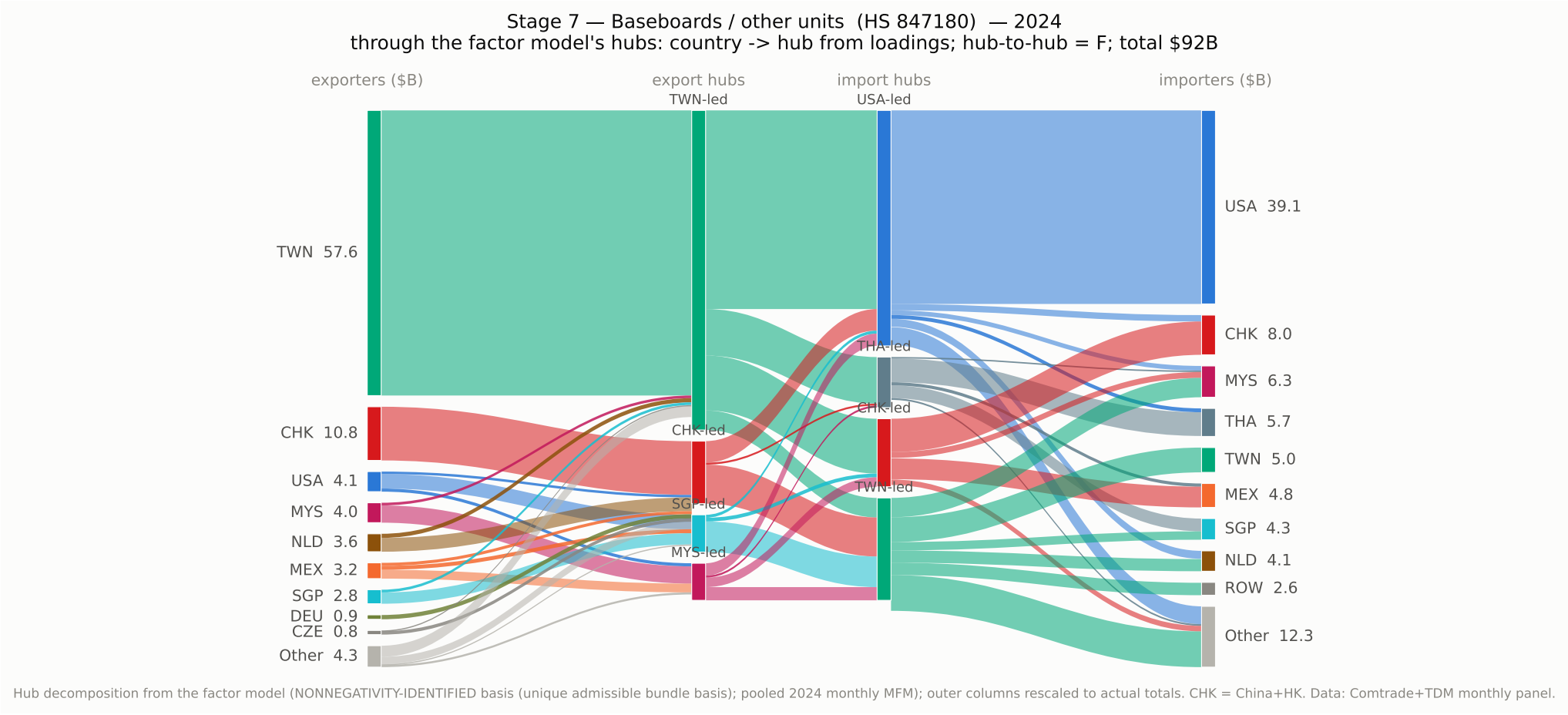
The biggest stage, and the whole electronics industry – phones, cars and PCs, not just AI. Most chips flow to Asian assembly; the China sphere absorbs about 43%.



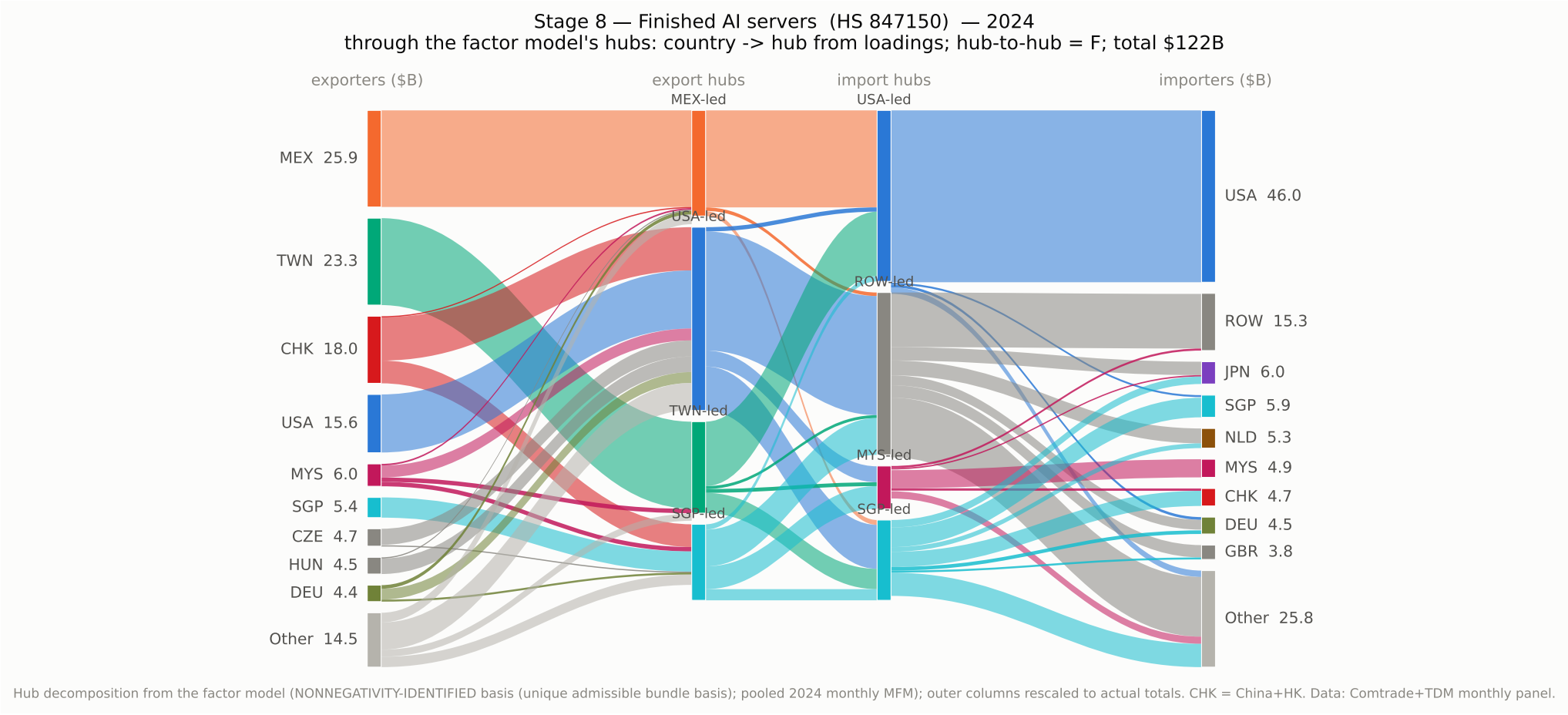
Stage 6 – parts and GPU modules (847330)



Stage 7 – baseboards (847180)



Stage 8 – AI servers (847150)



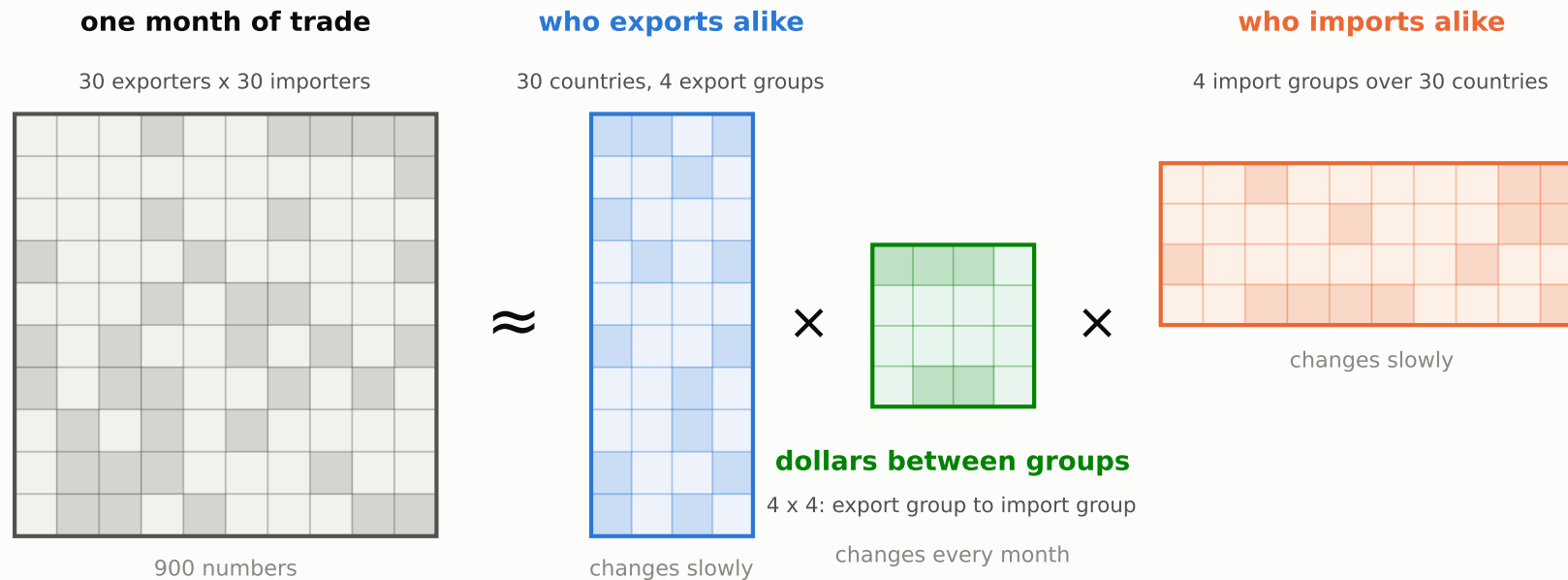
### 3. The factor model over time – AI-compute codes only

This section uses only the three blue codes (parts/GPU modules, baseboards, AI servers), monthly from 2017. Unless the page says otherwise, China and Hong Kong are counted as one. Shaded bands mark the periods between structural breaks.

## How the model works

The one idea behind every page that follows.

### How the factor model summarizes the trade data

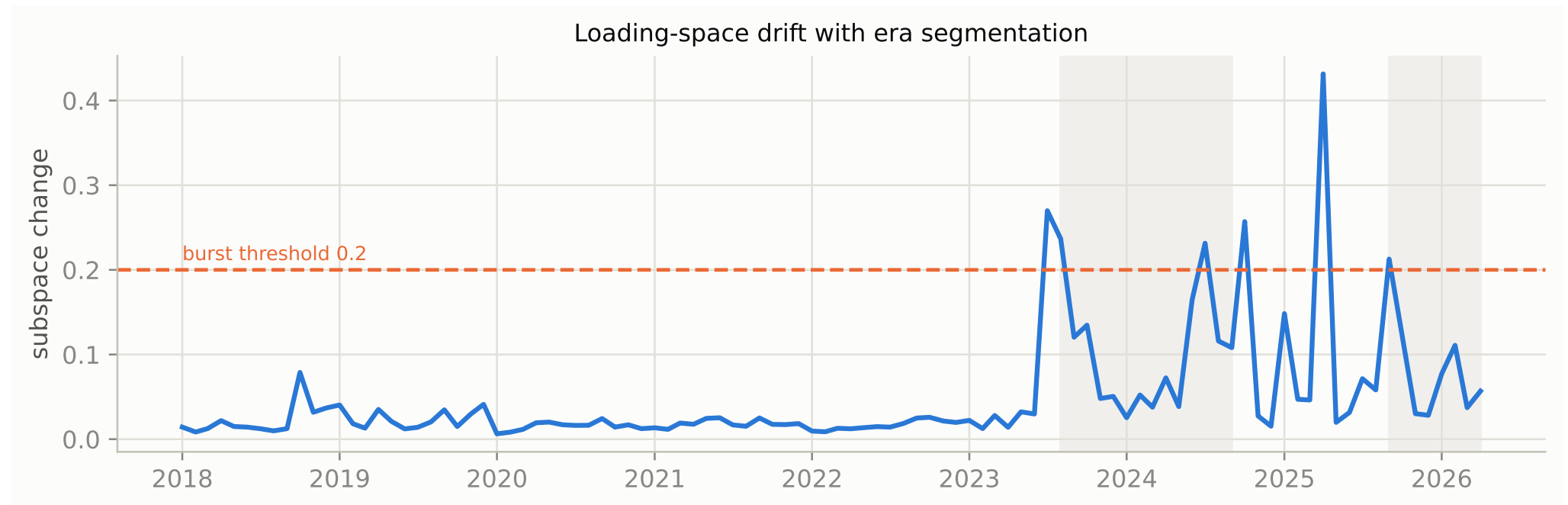


Every month, 900 country-to-country flows are compressed into: groups of exporters that move together, groups of importers, and a small table of dollars between the groups. The groups are the slow structure of the network -- when they shift, the model marks a break. The small table is the fast part: its cells are the trade channels, and three of them carry the AI boom.



## When the network changed shape

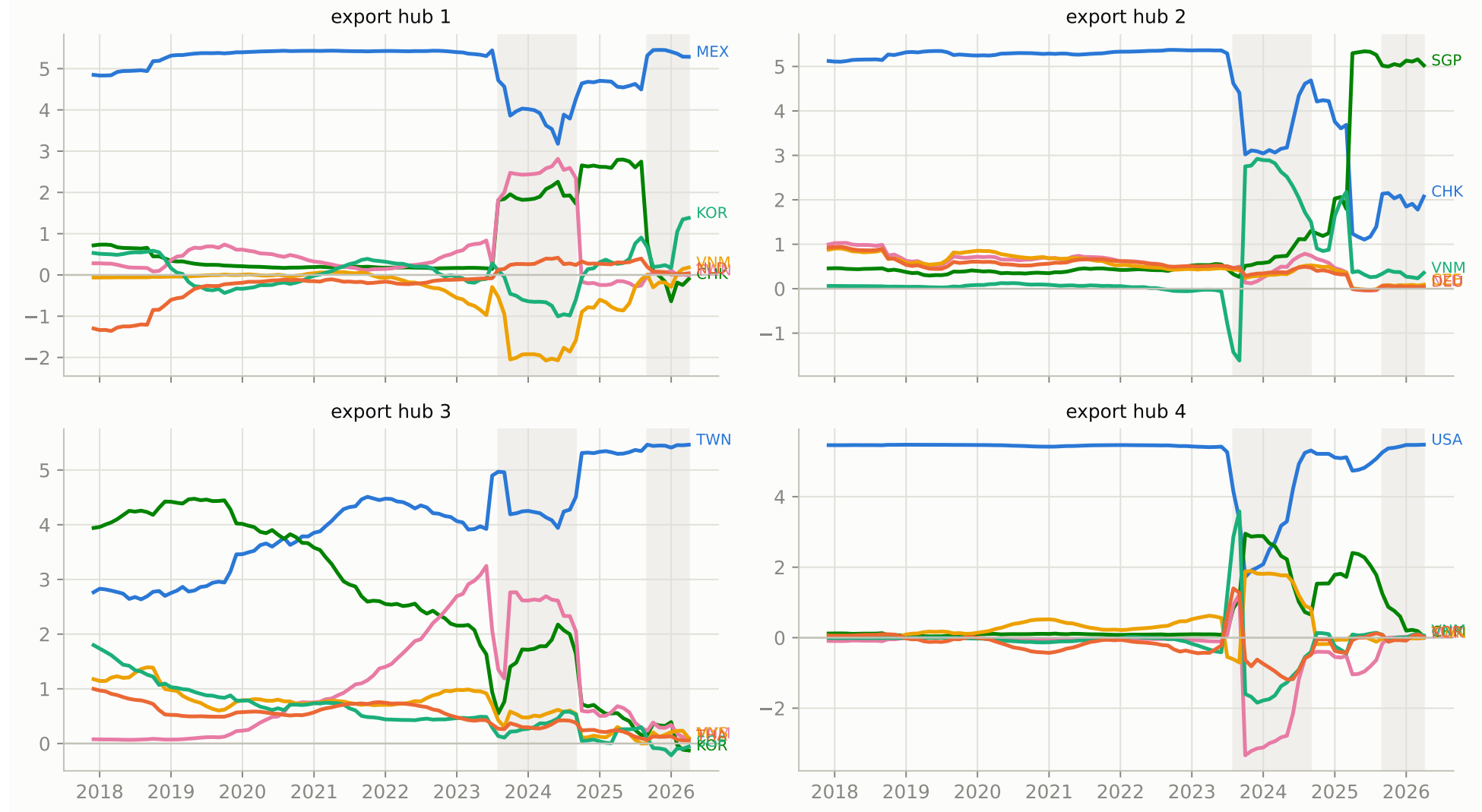
The line measures how much the export groupings moved from one month to the next. Flat for six years, then: Aug 2023 (AI demand), Oct 2024 (the quarter memory export controls were signaled), Sep 2025.



## Who is in each export group

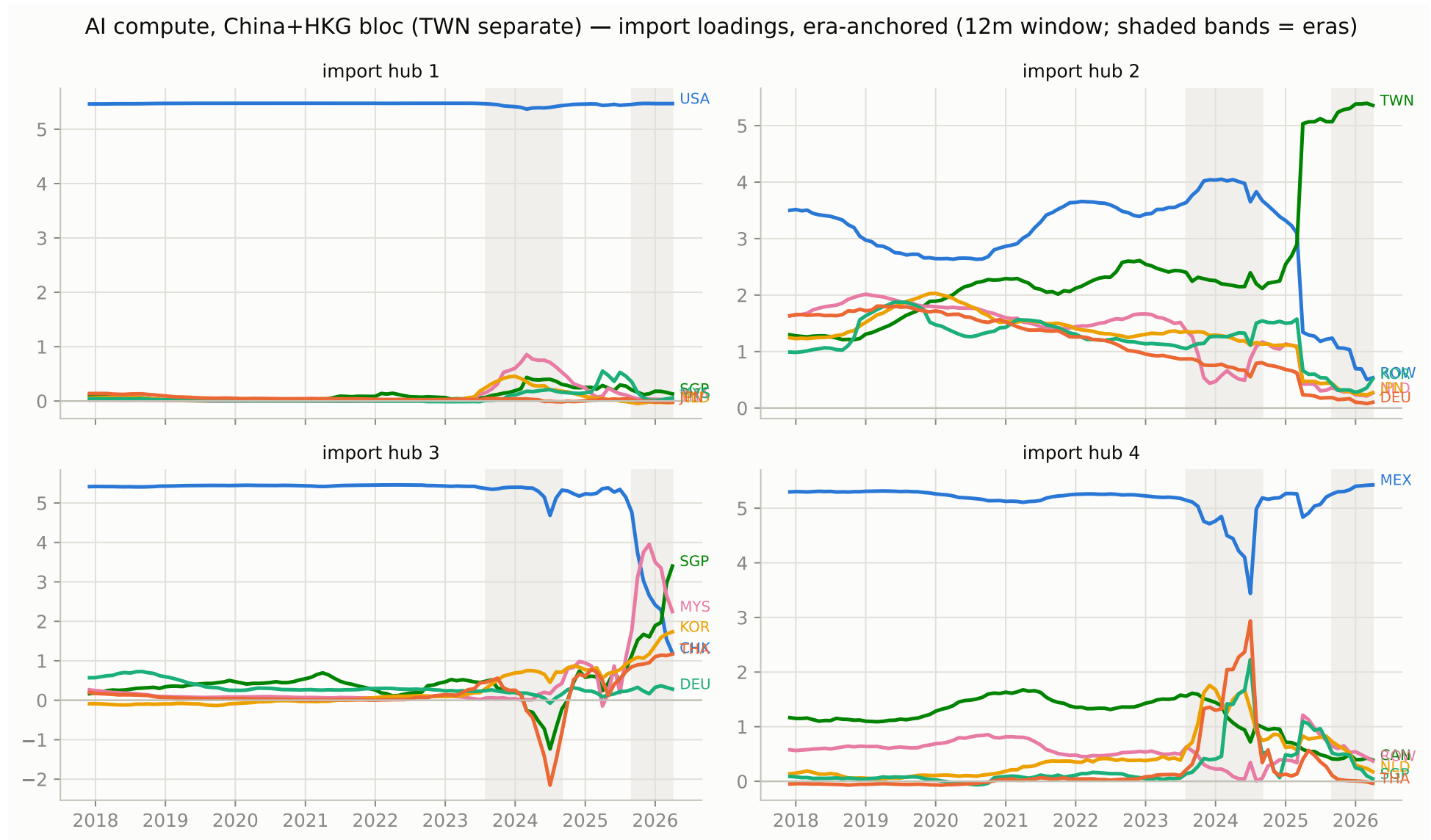
At Oct 2024 the China group loses its separate identity and a Singapore-led group appears.

AI compute, China+HKG bloc (TWN separate) — export loadings, era-anchored (12m window; shaded bands = eras)



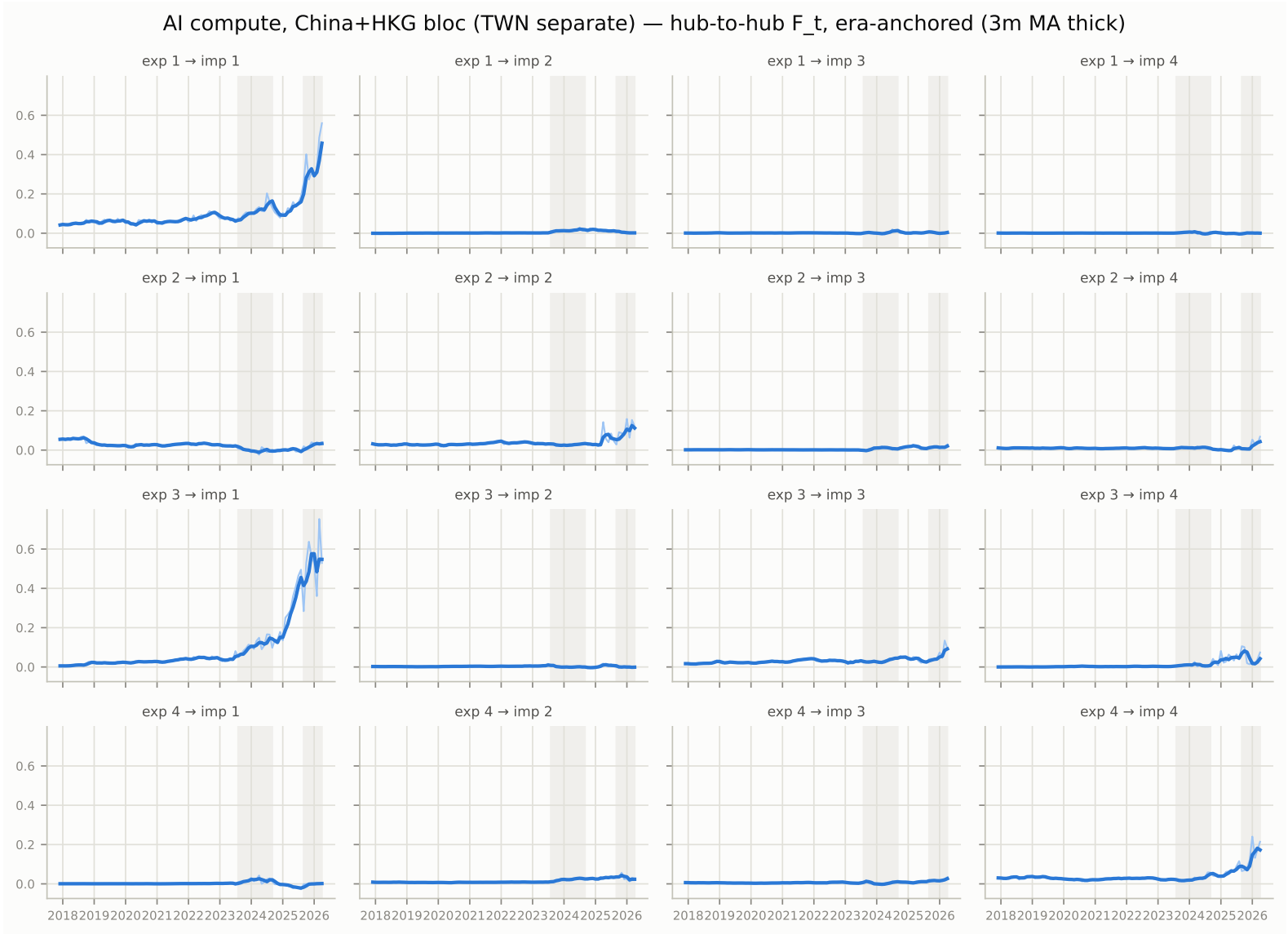
## Who is in each import group

The import side is steady: a US group, a Mexico group, a Taiwan group, and the rest of the world.



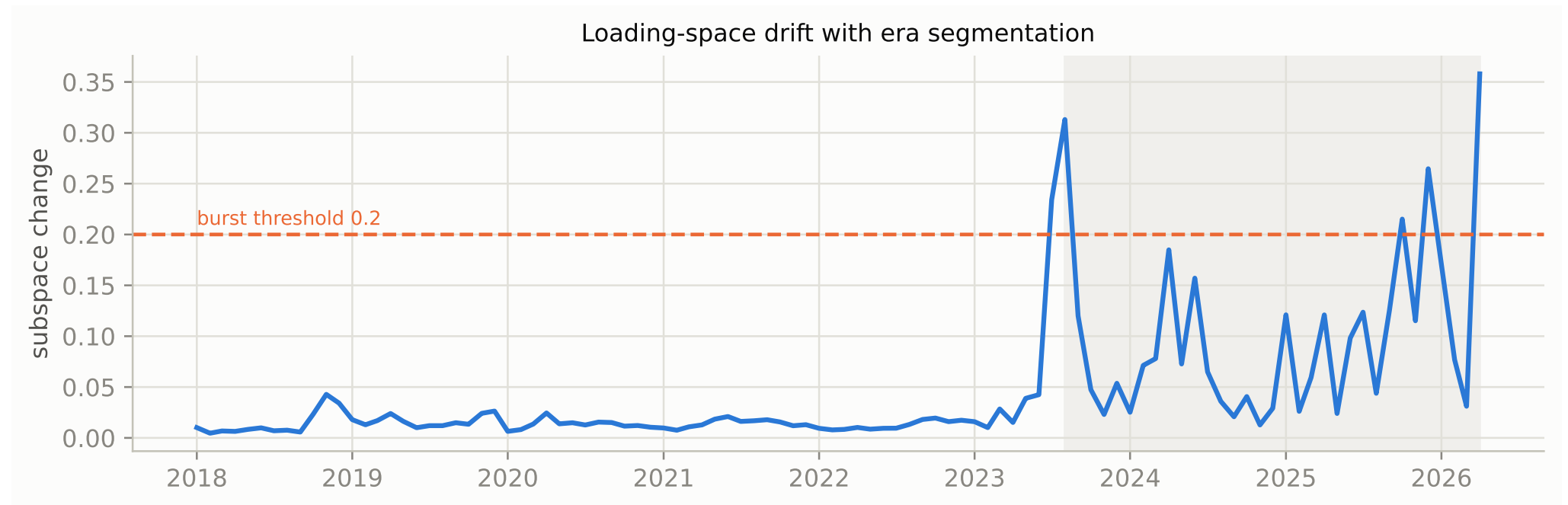
# Dollars between groups, month by month

Three of the sixteen channels carry the AI boom: Taiwan to the US, and both legs of the US-Mexico assembly loop. The rest are flat.



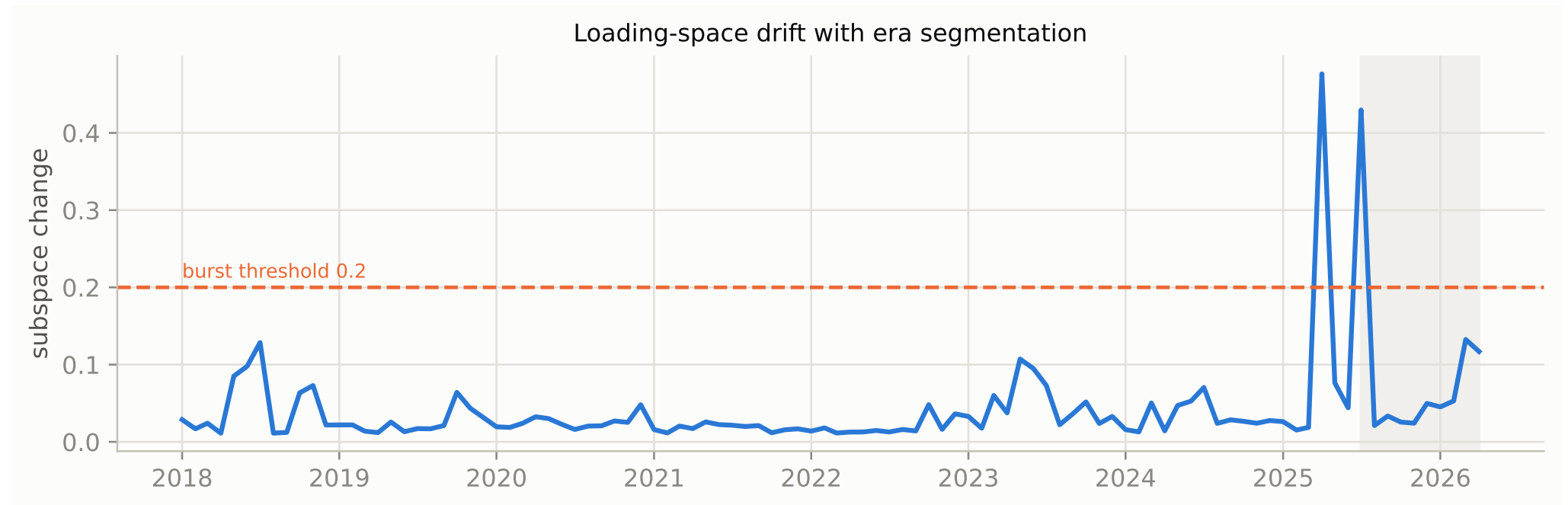
## Same, with every country separate

Without merging China and Hong Kong there is one break in nine years: Aug 2023. Demand changed the country-level structure; policy did not.



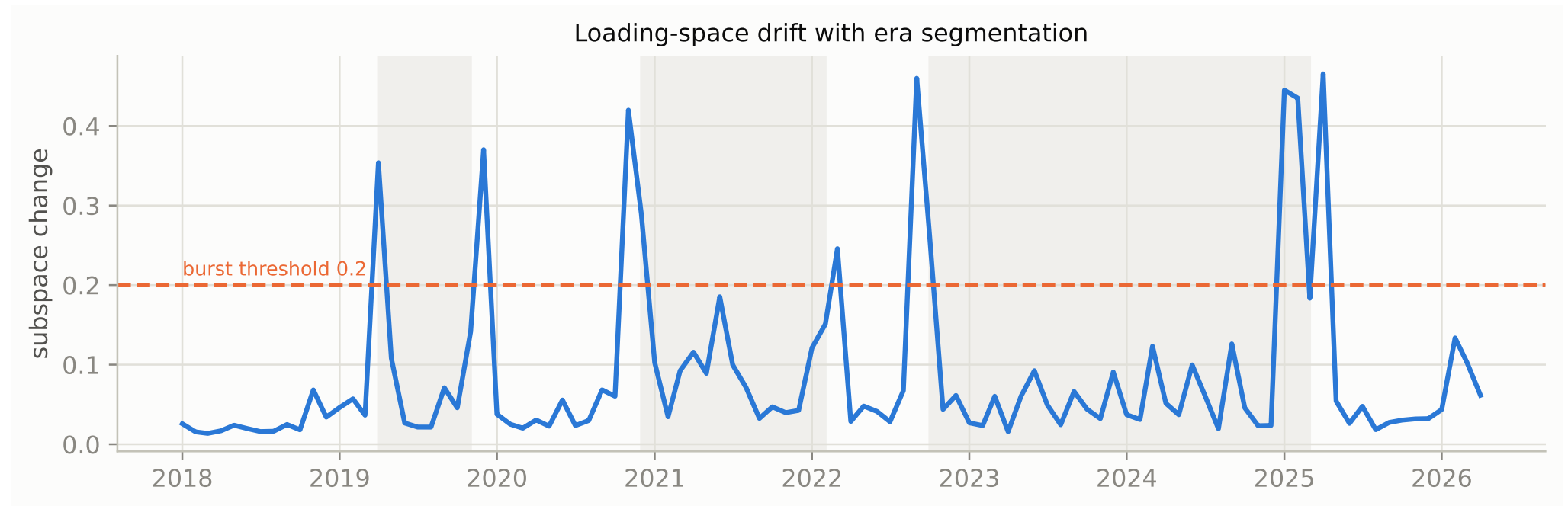
## Parts only (847330)

91 months without a break – the stable base layer of the network.



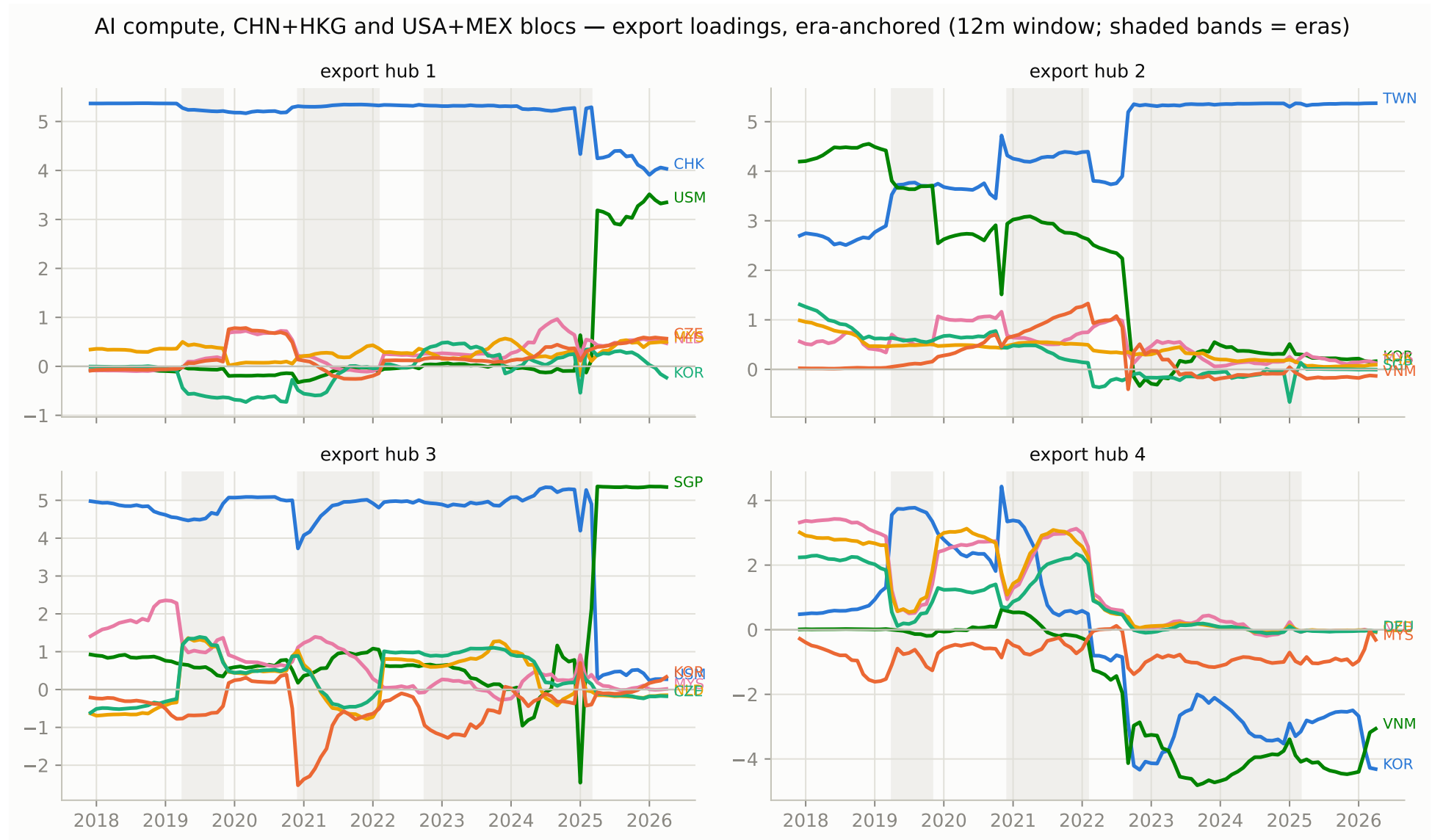
## With both blocs merged

Merging China+Hong Kong and US+Mexico hides churn inside each bloc and isolates what happens between them. The breaks line up with policy: the 2019 tariff war, the Oct 2022 export controls, the Apr 2025 tariffs.



## The tariff signature

After Apr 2025 the two blocs move together on a single factor – seen at no other break.





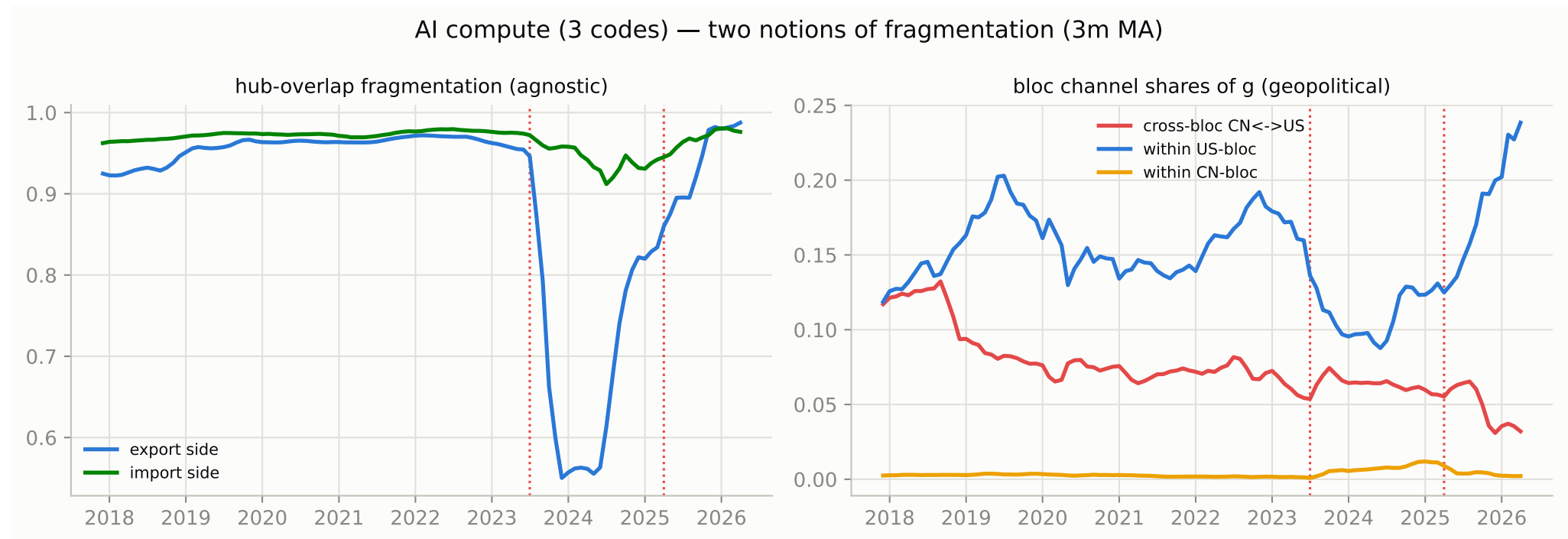
## 4. Concentration & fragmentation – AI-compute codes only

Monthly measures built on top of the model, for the three blue codes only. Dotted lines mark Jul 2023 and Apr 2025.

Interpretation: [results/network\\_stats/findings.md](#).

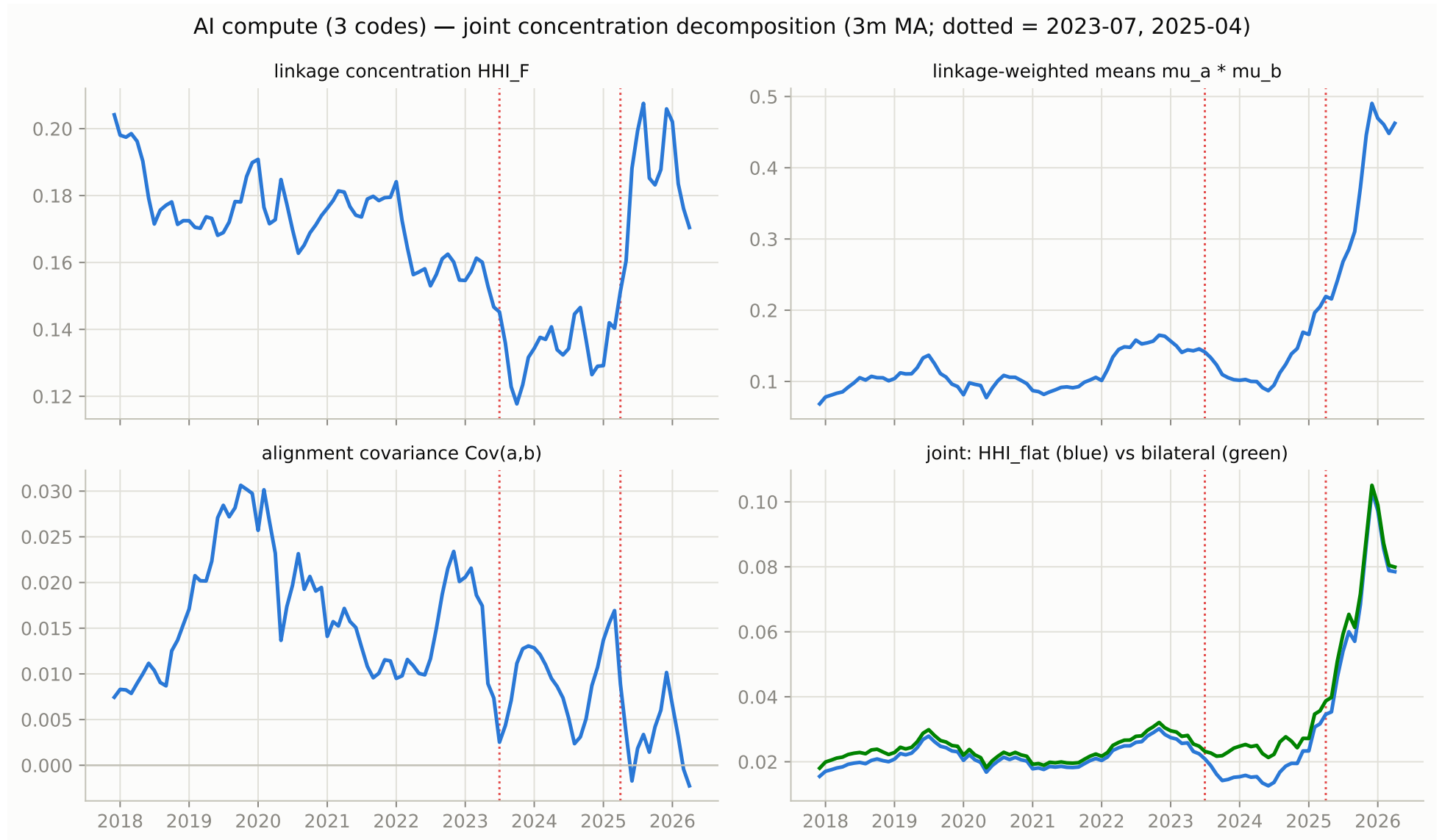
## Two meanings of fragmentation

Left: do the model's groups overlap (a technical check, little economic signal). Right: the share of trade crossing between the China and US blocs – down about 70% since 2017, in two steps (2019, then 2023 on), with no recovery between.



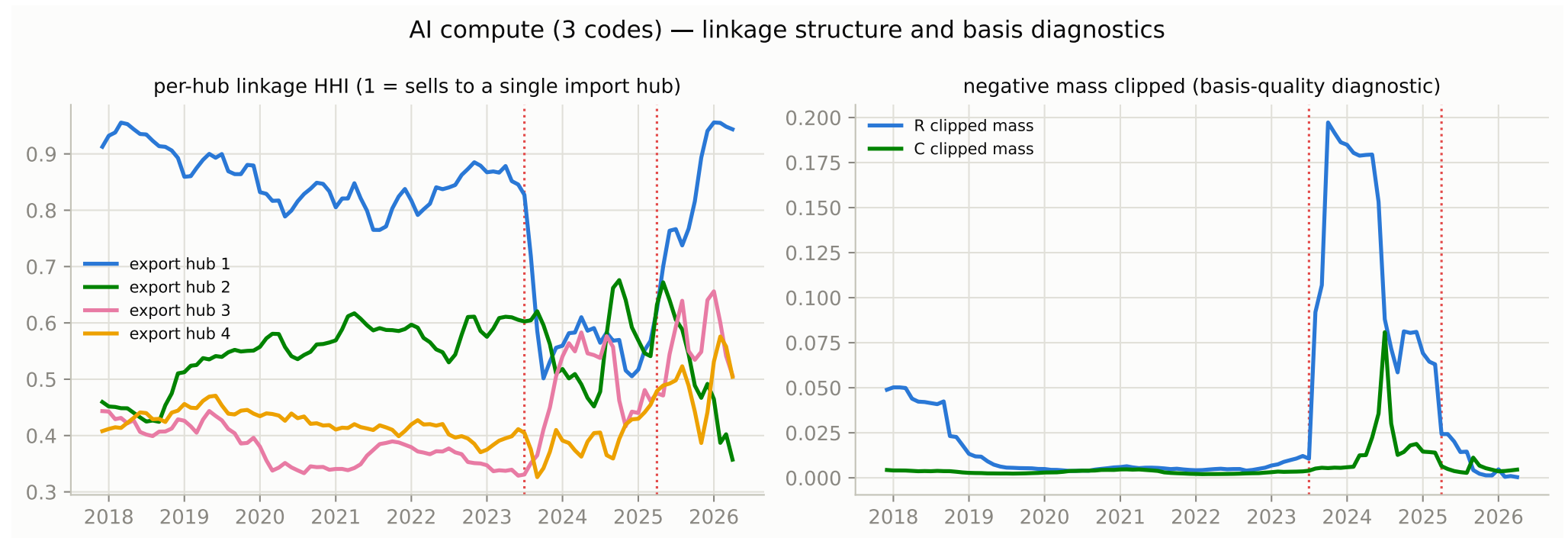
## Concentration, taken apart

Overall concentration splits exactly into three parts. The interesting one asks: do concentrated sellers ship to concentrated buyers? Mildly yes, peaking around 2019, fading through the AI era.



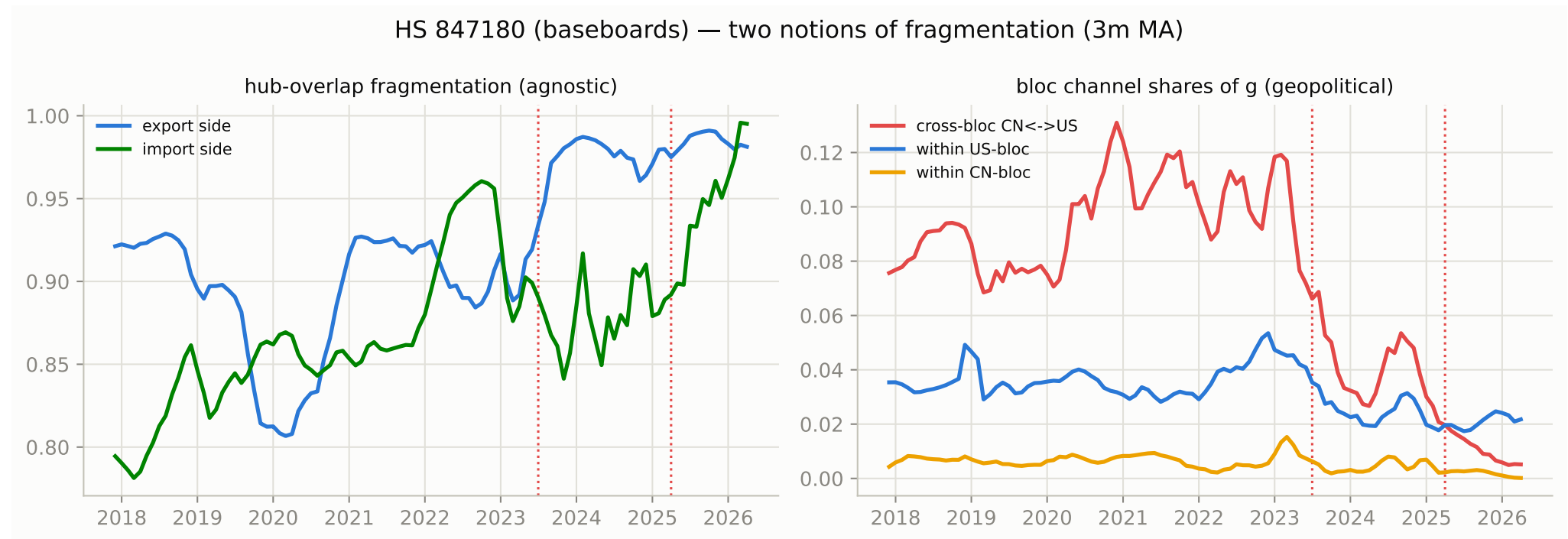
## One buyer or many?

Left: does each export group sell into one import group or several. Right: a data-quality check, elevated during the 2023-24 transition.



## Baseboards: the sharpest split

The cross-bloc share collapses about 20x from its 2021 peak – the code most exposed to export controls.



## Parts: still shared

The cross-bloc share halves but stays the highest of the three codes: the parts trade remains the layer both blocs still share.

